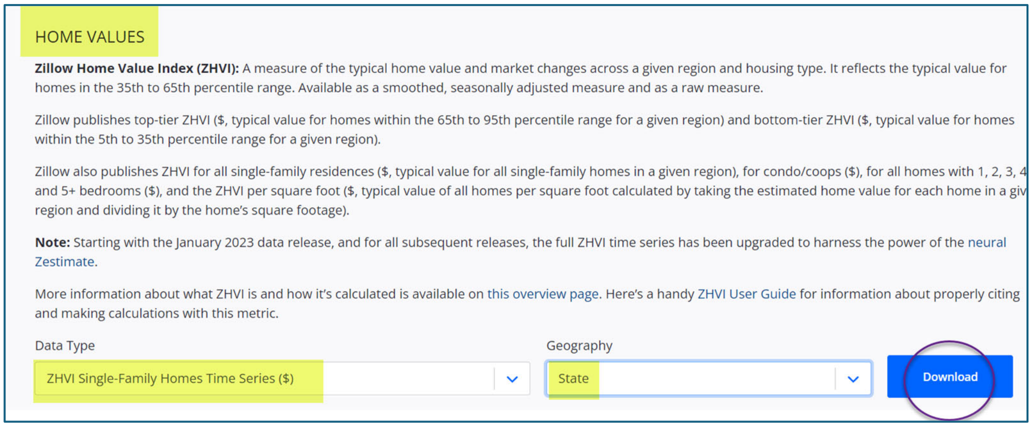
Practice with Pandas Dataframe Objects – Nyhoff

Get the Data

- As before, download US **State** data for **Single Family Homes** from Zillow:
  `https://www.zillow.com/research/data/`



- Compile the year-end data (**12/31/20yy**) dat for **your assigned state** for each of the past **10** years and format it like this in an Excel spreadsheet:

	A	B
1	YearEndData	AvgHomeValue
2	12/31/2015	$163,202.97
3	12/31/2016	$171,967.12
4	12/31/2017	$182,124.96
5	12/31/2018	$192,940.13
6	12/31/2019	$205,353.77
7	12/31/2020	$226,173.40
8	12/31/2021	$250,635.68
9	12/31/2022	$269,983.34
10	12/31/2023	$287,998.06
11	12/31/2024	$304,518.03

Data Analysis

Using this data, do the following in Pandas:

- Read this data into a **DataFrame** object named **xxTenYearEndDF**
  (where "**xx**" is the two-letter abbreviation for your assigned state).
- Then, using **subsetting**, pull out the column that contains
  the average home values and assign it to a series named **xxYearEndDatesSeries**
  (where "**xx**" is the two-letter abbreviation for your assigned state).
- In the console, display "**Year End Averages:**".
- In the console, print **xxYearEndDatesSeries**
- In the console, display "**Third year:**" followed by
  the year-end average home value for the third year.
  For this, use **iloc[]** notation.
- In the console, display "**Sixth year:**" followed by
  the year-end average home value for the sixth year.
  For this, use **iloc[]** notation.
- In the console, display "**Ninth year:**" followed by
  the year-end average home value for the sixth year.
  For this, use **iloc[]** notation.
- Write Pandas code to assign **new row** labels to the `DataFrame`.
  The row labels should have the following format: **SSyearEnd20yy**
  - **SS** is the two-letter abbreviation for your assigned state.
  - **yy** is the last two digits of that year

  Example for Wisconsin:

```
              YearEndData  AvgHomeValue
WIyearEnd2014  12/31/2014  $154,512.98
WIyearEnd2015  12/31/2015  $161,577.39
WIyearEnd2016  12/31/2016  $170,254.24
WIyearEnd2017  12/31/2017  $180,310.90
WIyearEnd2018  12/31/2018  $191,018.35
WIyearEnd2019  12/31/2019  $203,308.35
WIyearEnd2020  12/31/2020  $223,920.60
WIyearEnd2021  12/31/2021  $248,139.22
WIyearEnd2022  12/31/2022  $267,294.16
WIyearEnd2023  12/31/2023  $285,129.45
```

- Then, write Pandas code that uses `loc[]` to display the **first**, **fifth**, and **last** rows using the **new row labels**.
  Your output should be something like this:

```
First row:
YearEndData       12/31/2014
AvgHomeValue     $154,512.98
Name: WIyearEnd2014, dtype: object

Fifth row:
YearEndData       12/31/2018
AvgHomeValue     $191,018.35
Name: WIyearEnd2018, dtype: object

Last row:
YearEndData       12/31/2023
AvgHomeValue     $285,129.45
Name: WIyearEnd2023, dtype: object
```

Submit Your Code and Results

- Copy and paste your **Thonny code** into a Word document
- Copy and paste your **Console output** into the same Word document
- **Save** this Word file as a **PDF**.
- **Submit** this PDF via **Canvas**.